

Progress Report: Sprint 4

CS.028 Optimizing Research Software Codes - 20 November 2025

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Executive Summary

- Communicated with our project partner to address some concerns and request more data. Our project partner believes we have enough data to work with currently, so we will edit the existing files to manage the latency concerns when testing our implementation.
- We are continuing to analyze the original implementation, creating error handlers, and drafting ways to solve the latency issues.
- Our next steps are to break down `load_config()` to determine the root cause of the runtime issues and continue to propose solutions.

Progress vs Plan

Planned Item	Status	Links	Notes/Blockers	Plan Change
Implement function testing	Partial	N/A	Still researching how to optimize the functions	Extend to next sprint
Documentation draft	Complete	GitHub	Architecture.md added for rxn2db.py	None
Program flowchart analysis	Partial	N/A	Currently analyzing the program to determine how many times each function is called	None
Add error handling	Partial	N/A	Error handling added for executable path	None
Prototype optimized functions	Not done	N/A	Explore and experiment possible program optimizations	None
Implement initial optimizations	Not done	N/A	Implement first program update	None
Add more sample data input	Complete	N/A	Project partner does not have additional data	None

Evidence of Working Software

CI/Build: [GitHub URL](#)

Risk and Quality

Risk	Owner	Status	Concrete Actions
Code complexity	Team	Active	Modulating code sections, adding inline comments
Setup inconsistencies	Team	Resolved	Documented environment dependencies
Limited testing time	Team	Active	Prioritize optimizing the slowest functions
Poor documentation of function tests	Team	Active	Log pre-optimization function summaries and their expected outputs
Hidden dependencies between functions	Team	Active	Use trace debugging tools to observe the inner workings of the original program
Unclear success criteria	Team	Resolved	Log all implemented optimizations with time test results

Bug count: zero current known errors, extremely poor runtime efficiency.

Next Goals

- Analyze `load_config()` to understand why it is taking such a significant amount of the runtime. [REQ-002]
- Identify specific bottlenecks inside `load_config()` and begin planning improvements. [REQ-002]
- Improve code structure and formatting to decrease linting errors. [REQ-012]
- Add comments to increase readability of the code processes. [REQ-012]
- Test for runtime improvements after making changes to `load_config()`. [REQ-006]

Team Process Reflection

After identifying `load_config()` as the primary bottleneck we have a more narrowed scope on where large improvements to runtime can be made, which will aid us in our upcoming goal of planning specific potential optimizations in the code. We also noticed that the code base we inherited contained many linting errors and we aim to fix linting problems and increase code readability with inline comments describing functionality.

Individual Contributions

Kathryn Butler

- Architecture and Risks
 - System Context and High-Level Architecture
 - Interfaces/APIs and Data Contracts
 - Consolidated Risk Register
 - Traceability Aid
- Progress Report
 - Executive Summary
 - Progress vs Plan
 - Risk and Quality
- Created path error-handling functions ([PR#2](#))
- Created documentation of rnx2db.py architecture ([PR#3](#))

Michael Mcallister

- Established next goals
- Built and integrated function timer
- Identified primary bottleneck function as load_config()
- Consolidated team reflection

Joseph Schaab

- Outlined the implementation plan to use memory mapping
- Updated the testing document.
- Reviewed the current testing pool and added notes for where changes will be needed for future commits.